

Theoretical extensions for verified replay: optimization, admission, entropy, and survival

Companion research note for the ModeBench / ReplayMaxRL manuscript

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The four follow-up questions admit useful theoretical answers with distinct premises. This note gives complete derivations, identifies their literature basis, and states what evidence would be needed to apply them to the runs. These are applications and specializations of established mathematical tools; no foundational novelty is claimed. The note contains no new training results.

Question	Available theoretical result	Required application evidence
Optimizer	Simultaneous high-probability retention from a controlled stochastic energy budget	Bounds on update bias, noise, smoothness, and preconditioner dependence
Admission	Adaptive hazard bounds plus retention across changes in the bank	Actual insertion opportunities, capacity, persistent coefficients, and switching costs
Exact entropy	Same full-support optimum as replay under conditional entropy, but different KL geometry	An exact estimator and explicit output alphabet; no transfer from a historical score
Per-mode survival	Sharp probability certificates and valid finite-count intervals	Complete-response likelihoods or per-key sampling counts under a specified decoding law

1 An optimizer-aware conditional guarantee

Stochastic retention with a controlled energy budget. The argument combines a standard variable-metric descent estimate (Wang et al., 2017, Section 2) with the nonnegative-supermartingale maximal inequality (Howard et al., 2020, Lemma 1). The probability consequence for protected exemplars is derived here; it is not an existing theorem about Adam or language-model replay.

Theorem 1 (Optimizer-aware conditional exemplar retention). *Fix finitely many complete verified responses (x_j, e_j) with keys b_j , lengths $L_j > 0$, and weights $\alpha_j > 0$. Let*

$$R(\theta) = \sum_j \frac{\alpha_j}{L_j} [-\log \pi_\theta(e_j \mid x_j)], \quad F(\theta) = R(\theta) - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

Assume F is differentiable with globally L_F -Lipschitz gradient, and take deterministic θ_0 with finite $F(\theta_0)$. Relative to a filtration $(\mathcal{F}_t)$, consider

$$\theta_{t+1} = \theta_t - \eta_t H_t(g_t + b_t + \xi_t), \quad g_t = \nabla F(\theta_t).$$

Here $\theta_t, \eta_t, H_t, b_t$ are $\mathcal{F}_t$ -measurable, $0 \preceq H_t \preceq MI$, $\eta_t > 0$, $\eta_t L_F M \leq 1$, and $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ with finite conditional second moment. Suppose deterministic numbers $d_t \geq 0$ bound

$$\frac{\eta_t}{2} \|b_t\|_{H_t}^2 + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 \mid \mathcal{F}_t] \leq d_t, \quad \|u\|_H^2 = u^\top H u.$$

For $N \in \mathbb{N} \cup \{\infty\}$ with $B_N = \sum_{t < N} d_t < \infty$, put $D_0 = F(\theta_0) + J_{\max} \geq 0$. For every $0 < \delta < 1$,

$$\Pr\left(\forall t \leq N, \forall j : p_{\theta_t}(b_j \mid x_j) \geq \pi_{\theta_t}(e_j \mid x_j) \geq \exp\left[-\frac{L_j(D_0 + B_N)}{\alpha_j \delta}\right]\right) \geq 1 - \delta.$$

When $N = \infty$, each protected probability also has an almost surely positive, potentially random, infimum over all iterations.

Proof. Smoothness, conditional centering of ξ_t , and $H_t^2 \preceq M H_t$ give

$$\begin{aligned} \mathbb{E}[F(\theta_{t+1}) \mid \mathcal{F}_t] &\leq F(\theta_t) - \eta_t g_t^\top H_t (g_t + b_t) + \frac{\eta_t}{2} \|g_t + b_t\|_{H_t}^2 + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 \mid \mathcal{F}_t] \\ &\leq F(\theta_t) - \frac{\eta_t}{2} \|g_t\|_{H_t}^2 + d_t. \end{aligned}$$

The cross term involving b_t cancels on expanding the square. Since $F + J_{\max} \geq R \geq 0$, the adapted process

$$X_t = F(\theta_t) + J_{\max} + \sum_{s=t}^{N-1} d_s$$

is a nonnegative supermartingale; for $N = \infty$ use the infinite tail. Its initial value is $D_0 + B_N$. The maximal inequality yields $\Pr(\sup_{t \leq N} X_t \geq (D_0 + B_N)/\delta) \leq \delta$. If $D_0 + B_N = 0$, nonnegativity instead gives $X_t = 0$ almost surely. Every nonnegative summand of $R(\theta_t) \leq X_t$ obeys the same bound, and exponentiation proves the simultaneous exemplar floors without a union factor over j . Each complete-exemplar event is contained in its verified-key event. For $N = \infty$, the same maximal inequality, with a threshold tending to infinity, gives $\sup_t X_t < \infty$ almost surely. $\square$

Finite and infinite horizons are different. Unbiased bounded-variance updates have $d_t = O(\eta_t^2)$; square-summable steps therefore suffice for the infinite-horizon energy budget. A persistent bias instead costs $O(\eta_t \|b_t\|^2)$. Constant steps and persistent variance give only the stated finite-horizon bound unless additional structure controls accumulated noise. No positive lower eigenvalue of H_t is needed for retention alone; this theorem does not assert parameter convergence or optimality. The likelihood must include the complete response and termination event under the sampling law whose key probability is bounded.

Scope for adaptive implementations. A current-gradient Adam second moment is not predictable with respect to that gradient's noise, and momentum, clipping, bank changes and cyclic prompt scheduling need separate error or energy control. Positive replay weight and a small nominal learning rate do not verify these assumptions. Reddi et al. (2018) provide counterexamples to unrestricted Adam convergence; this does not itself prove replay failure in the present application. The bound above is conditional and can be numerically weak, particularly for long exemplars or small replay weights.

2 From actual admission to persistent protection

Conditional Borel–Cantelli supplies the standard probability background [6, Theorem 4.3.4]. Controlling energy across switches has a long history in hybrid-system analysis [7]; the elementary positive-jump accounting below is proved directly. Archive methods such as Go-Explore separate discovery, return, and policy learning [8]. That is a conceptual precedent, not a theorem about the present response bank.

Discovery requires admission opportunities. The following is a standard conditional-hazard argument, closely related to conditional Borel–Cantelli (Durrett, 2019, Theorem 4.3.4), applied to insertion of a verified key into memory. Its hazard includes scheduling, validation, capacity, and candidate selection; a positive probability of generating a key alone is insufficient.

Proposition 2 (Conditional admission hazards). *Fix a key b not initially banked. Let $(\mathcal{F}_n)_{n \geq 0}$ describe the complete training history after each opportunity, and let τ_b be its first admission time, with $\tau_b = \infty$ if it is never admitted. Define the predictable stopped hazard*

$$h_{b,n} = \Pr(\tau_b = n \mid \mathcal{F}_{n-1}), \quad H_{b,N} = \sum_{n=1}^N h_{b,n}.$$

For every $a \geq 0$,

$$\Pr(\tau_b > N, H_{b,N} \geq a) \leq e^{-a}.$$

In particular, $\Pr(\tau_b = \infty, H_{b,\infty} = \infty) = 0$. If deterministic numbers $\underline{h}_{b,n} \in [0, 1]$ satisfy $h_{b,n} \geq \underline{h}_{b,n}$ on $\{\tau_b \geq n\}$, then

$$\Pr(\tau_b > N) \leq \prod_{n=1}^N (1 - \underline{h}_{b,n}) \leq \exp\left(-\sum_{n=1}^N \underline{h}_{b,n}\right).$$

For a finite target set S , the probability that some target key has not been admitted by N is at most the sum of these bounds over $b \in S$.

Proof. Let $Z_N = \mathbf{1}\{\tau_b > N\} \exp(H_{b,N})$, with $Z_0 = 1$. The event of an earlier admission is known at time $n - 1$, and $h_{b,n} = 0$ on that event. On survival through $n - 1$,

$$\mathbb{E}[Z_n \mid \mathcal{F}_{n-1}] = Z_{n-1} e^{h_{b,n}} (1 - h_{b,n}) \leq Z_{n-1}.$$

Thus Z_n is a nonnegative supermartingale. Markov’s inequality gives the finite-hazard bound. Fatou’s lemma excludes a positive-probability event on which survival continues forever while $H_{b,N} \rightarrow \infty$, since then $Z_N \rightarrow \infty$. Under deterministic hazard floors, conditioning gives $\Pr(\tau_b > n) \leq (1 - \underline{h}_{b,n}) \Pr(\tau_b > n - 1)$; iteration and the union bound finish the proof. $\square$

Divergent cumulative hazard is a substantive premise. Independent opportunities with admission probabilities $h_n = (n + 1)^{-2} > 0$ satisfy

$$\Pr(\tau = \infty) = \prod_{n=1}^{\infty} \left(1 - \frac{1}{(n + 1)^2}\right) = \frac{1}{2}.$$

Likewise, once a non-evicting capacity- K bank is full, every unbanked key has zero admission hazard. These bounds establish coverage only for keys that continue to have admissible opportunities; they do not establish such opportunities for the implemented policy or for supports larger than capacity.

Retention across bank changes. An admitted key also needs persistent loss weight. The next elementary energy accounting result allows bank changes while explicitly charging their effect on the objective. It is not an assumption that different bank objectives share a single unchanged potential. This follows the standard switched-energy viewpoint (Branicky, 1998); the specific inequality below is proved directly.

Proposition 3 (Switching-energy retention). *Let $t_0 < t_1 < \dots$ be switching times with finitely many switches on every bounded time interval. On interval r , use a finite collection of verified complete-response exemplars, with nonnegative coefficients d_{rj} and*

$$F_r(\theta) = \sum_{j:d_{rj}>0} d_{rj}\ell_j(\theta) - J(\theta), \quad \ell_j(\theta) = -\log \pi_\theta(e_j | x_j), \quad J(\theta) \leq J_{\max} < \infty.$$

Zero coefficients denote absent exemplars and are omitted from the sum. Suppose initial energy is finite, parameters are continuous at switches, F_r is nonincreasing within its interval, and the positive energy jumps obey

$$A := \sum_{r \geq 1} [F_r(\theta(t_r)) - F_{r-1}(\theta(t_r))]_+ < \infty.$$

If exemplar j is protected from interval r_j onward and $d_{rj} \geq \underline{d}_j > 0$ for every $r \geq r_j$, then, with $D = F_0(\theta(t_0)) + A + J_{\max} < \infty$,

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp(-D/\underline{d}_j) \quad (t \geq t_{r_j}).$$

The likelihood must represent the complete verified response under the law whose key probability is bounded. Mean-token losses have $d_{rj} = \alpha_{rj}/L_j$ for fixed exemplar length $L_j > 0$.

Proof. Induction over the switches gives $F_r(\theta(t)) \leq F_0(\theta(t_0)) + A$. Therefore $\sum_j d_{rj}\ell_j(\theta(t)) \leq D$. Each summand is nonnegative, so a protected exemplar has $\ell_j(\theta(t)) \leq D/\underline{d}_j$. Its complete response event is contained in its verified-key event. $\square$

If old coefficients are preserved when new exemplars are added, a switch costs exactly the summed weighted surprisal of the new exemplars at admission. For uniform categorical replay with fixed dose ρ , adding one key to a size- k bank instead changes the potential by

$$\Delta F = \frac{\rho}{k+1} [-\log p_{\text{new}} - R_{\text{old}}].$$

Old coefficients decrease from ρ/k to $\rho/(k+1)$, but remain at least ρ/K when capacity is K . Finitely many such admissions, with positive admission-time probabilities and no eviction, therefore have finite total switching cost in the exact descent model. Recurrent scheduling, bounded priority multipliers, or a finite-capacity buffer alone do not establish within-interval descent or a summable jump budget for arbitrary optimization.

Corollary 4 (Conditional discovery followed by retention). *Suppose a prompt has a finite correct support $\mathcal{C}$ of size m no larger than bank capacity. Admissions are non-evicting, and every missing key b has conditional admission hazard at least ϵr_b at every opportunity, for constants $\epsilon > 0$ and $r_b > 0$ with $\epsilon r_b \leq 1$. Assume also the switching-energy conditions above, including a finite positive-jump budget and persistent positive coefficients for the admitted complete exemplars. Then the bank reaches full correct coverage in finitely many opportunities almost surely, and thereafter every correct key has a uniform-in-time positive probability floor. At horizon N ,*

$$\Pr(\text{full coverage by } N) \geq 1 - \sum_{b \in \mathcal{C}} e^{-N\epsilon r_b}.$$

If $r_b \geq \gamma > 0$ for every correct key, this is at least $1 - me^{-N\epsilon\gamma}$.

Proof. Apply the admission-hazard bound to each initially missing key and take a union bound. The failure bound tends to zero; finite support and non-eviction therefore imply a finite almost-sure opportunity index of simultaneous full coverage. The switching-energy bound protects each admitted complete exemplar and hence its key, and applies to all keys after that time. $\square$

A fixed full-support proposal mixture supplies such a hazard floor if it is queried at every counted opportunity and insertion is guaranteed whenever the key appears. This is a sufficient design condition, not a property established for the current temperature schedule, bounded search attempts, or adaptive priority controller. The corollary guarantees retention, not uniform key probabilities or convergence of the neural optimizer.

3 Exact entropy and replay: the same target can have different protection

The KL-direction distinction is established in the label-smoothing literature [1, Section 3.2]. The corresponding log-barrier regularizer is also analyzed for policy gradient [2, Section 5.2]. Our specialization concerns a target restricted to verified modes, rather than uniform regularization over every action. A stored bank supplies a target that can be scored even when fresh sampling misses it.

Let $\mathcal{C}$ contain $m \geq 2$ correct categories, let $P = \sum_{c \in \mathcal{C}} p_c > 0$, $q_c = p_c/P$, and let $u_c = 1/m$. Write $H(q) = -\sum_c q_c \log q_c$, with $0 \log 0 = 0$.

Proposition 5 (Full-support objective comparison). *For a full correct-mode bank with a uniform categorical target,*

$$H(q) = \log m - \text{KL}(q||u), \quad R_u(p) := -\frac{1}{m} \sum_{c \in \mathcal{C}} \log p_c = \log m - \log P + \text{KL}(u||q).$$

If Ψ is continuous and strictly increasing on $[0, 1]$, then for any $\beta, \rho > 0$, both

$$\Psi(P) + \beta H(q) \quad \text{and} \quad \Psi(P) - \rho R_u(p)$$

have the unique maximizing distribution $P = 1, q = u$ on the closed simplex (with the conditional-entropy objective considered on $P > 0$). This is a statement about global maximizers, not about convergence of stochastic updates or conditional-entropy gradient flow.

Proof. Substitute $p_c = Pq_c$ into R_u , and expand both KL divergences. Since $H(q) \leq \log m$, equality only at $q = u$, maximizing the first objective selects $q = u$ and then $P = 1$. The second is $\Psi(P) + \rho \log P - \rho \log m - \rho \text{KL}(u||q)$; its first two terms increase strictly with P , while the last is maximized only at $q = u$. At that point there is zero incorrect mass, so the distribution is unique. $\square$

Full-output entropy is a different comparator. For a linear correctness potential aP , exact maximization of $aP + \beta H(p)$ gives

$$p_i^* = \frac{\exp(a \mathbf{1}\{i \in \mathcal{C}\}/\beta)}{m \exp(a/\beta) + n},$$

where $n \geq 1$ is the number of incorrect categories. This is the standard entropy/conjugacy calculation [3, Section 2.2]. It has $P^* < 1$, whereas exact conditional entropy has no such loss of correctness at its global optimum. The experimental historical semantic score is not an exact gradient of either objective.

Existing discrete-time theory is stronger than merely finding an optimum: Mei et al. prove positive action-probability floors and geometric convergence for their exact tabular entropy-regularized policy-gradient update [4, Lemma 13 and Theorem 5]. This provides relevant prior theory for an exact finite categorical comparator, not for the historical predictor, clipped PPO, or AdamW. For the paper’s Dr.GRPO reduction the reward vector is $a\mathbf{1}_C$ with $a = (G - 1)/G \in (0, 1)$, so their bandit reward assumption applies. Their linear-reward theorem does not directly cover the nonlinear finite-group MaxRL potential.

The converse directions of KL explain a certificate difference: $\text{KL}(u\|q)$ diverges if any protected coordinate vanishes, while $\text{KL}(q\|u)$ remains finite. For example, losing one mode and distributing mass uniformly over the rest leaves $H(q) = \log(m - 1)$. Section 4 quantifies this sharply. This level-set fact does not imply that exact entropy dynamics necessarily lose a mode.

4 Sharper per-mode certificates and measurement limits

This section applies standard relative-entropy data processing, binary-divergence inversion, and binomial inference to replay retention. The information-theory facts are established results [9]; binary-KL inversion is also a standard tool in statistical learning [10]. The propositions below give self-contained specializations, not claims of new foundational inequalities. They distinguish deterministic probability certificates from sampling confidence and from training-trajectory guarantees. All logarithms are natural. No claim here establishes energy descent for a stochastic neural optimizer.

4.1 A sharp certificate from normalized replay cross entropy

For probability vectors, write $H(w) = -\sum_i w_i \log w_i$ and

$$\text{kl}(a, r) = a \log \frac{a}{r} + (1 - a) \log \frac{1 - a}{1 - r},$$

with the usual continuous endpoint conventions. For $0 < a < 1$ and $D \geq 0$, define $\ell_-(a, D)$ as the unique $r \in (0, a]$ satisfying $\text{kl}(a, r) = D$. Define $\ell_-(1, D) = e^{-D}$.

Proposition 6 (Sharp replay certificate). *Let $w_b > 0$, $\sum_{b \in B} w_b = 1$, and let p be a distribution on an alphabet containing the bank B . Suppose*

$$R_w(p) = -\sum_{b \in B} w_b \log p_b \leq C < \infty, \quad D = C - H(w).$$

Then $D \geq 0$ and every banked coordinate satisfies

$$p_b \geq \ell_-(w_b, D) > 0.$$

The bound is sharp on the closed probability simplex. Furthermore,

$$p(B) \geq e^{-D}, \quad p_b \geq \max \left\{ 0, w_b - \sqrt{D/2} \right\}.$$

For $w_b < 1$ a simple positive lower bound, weaker than binary inversion, is $\exp[-(D + h_2(w_b))/w_b]$, where $h_2(a) = H((a, 1 - a))$.

Proof. Extend w by zero off the bank, writing $\bar{w}$. Normalization gives $\text{KL}(\bar{w}\|p) = R_w(p) - H(w) \leq D$. Put $a = w_b$ and $r = p_b$. The log-sum inequality on all other categories gives

$$\sum_{j \neq b} \bar{w}_j \log \frac{\bar{w}_j}{p_j} \geq (1 - a) \log \frac{1 - a}{1 - r},$$

so $\text{kl}(a, r) \leq D$. This is data processing under the binary partition $\{b\}, \{b\}^c$. On $r \in (0, a)$ the derivative of $\text{kl}(a, r)$ is $(r-a)/(r(1-r)) < 0$; the divergence tends to infinity at zero and equals zero at $r = a$. This proves the inverse bound. Equality is attained by taking $p_b = r$ and $p_j = (1-r)w_j/(1-a)$ on the other banked categories, with zero mass off the bank. If strict full support is required, this is the limiting construction; the infimum is approached as off-bank mass vanishes. The case $a = 1$ is immediate.

Writing $s = p(B)$ and $\tilde{p}_b = p_b/s$ gives $\text{KL}(\bar{w}||p) = -\log s + \text{KL}(w||\tilde{p}) \geq -\log s$. For the coordinate Pinsker bound, fix r and differentiate the binary KL twice in its first argument: $\partial_a^2 \text{kl}(a, r) = 1/[a(1-a)] \geq 4$. Its value and first derivative vanish at $a = r$, so $\text{kl}(a, r) \geq 2(a-r)^2$. Finally, $\text{kl}(a, r) = -h_2(a) - a \log r - (1-a) \log(1-r) \geq -h_2(a) - a \log r$, giving the elementary exponential bound. $\square$

For the paper’s energy inequality, substitute $C = R_w(p(T)) + [\Psi(1) - \Psi(P(T))]/\rho_w$. This improves the existing termwise bound $p_b \geq e^{-C/w_b}$ because it uses probability normalization and subtracts target entropy. It can also be used at an individual checkpoint if a genuine upper bound on its complete bank cross entropy is available; a checkpoint calculation does not prove that the same bound holds at later times.

For a uniform bank of 16, $C = \log 16 + 0.01$ certifies $p_b \geq 0.0339712$, whereas the termwise bound is approximately 4.62×10^{-20} . These are illustrative mathematical inputs, not measured training losses. At excess cross entropy $D = 160$, even the sharp log lower bound is about -2563.74 , so normalization cannot rescue an intrinsically enormous energy budget.

4.2 What a scalar entropy value can certify

Proposition 7 (Sharp entropy threshold for all-mode retention). *Let q be a probability vector on exactly $m \geq 2$ possible modes. A scalar bound $H(q) \geq h_0$, with $h_0 \leq \log m$, gives a strictly positive uniform coordinate floor if and only if $h_0 > \log(m-1)$. In that case*

$$q_b \geq r_m(h_0) \quad \text{for every } b,$$

where $r_m(h_0) \in (0, 1/m]$ uniquely solves

$$h_2(r) + (1-r) \log(m-1) = h_0.$$

The certificate is sharp.

Proof. Fix $r = q_b$. Splitting off that mode gives

$$H(q) = h_2(r) + (1-r)H(q_{-b}/(1-r)) \leq h_2(r) + (1-r) \log(m-1) =: f_m(r).$$

Equality is attained when the other modes are uniform. On $0 < r < 1/m$, $f'_m(r) = \log[(1-r)/(r(m-1))] > 0$, with $f_m(0) = \log(m-1)$ and $f_m(1/m) = \log m$. If $H(q) \geq h_0 > \log(m-1)$, a coordinate below the stated root is impossible. Conversely, if $h_0 \leq \log(m-1)$, a distribution with one zero coordinate and the rest uniform satisfies the bound. Even restricting to positive coordinates permits arbitrarily small values by approximation. $\square$

For uniform u , $H(q) = \log m - \text{KL}(q||u)$, so the same certificate follows from $\text{kl}(q_b, 1/m) \leq \text{KL}(q||u)$. It is positive precisely when the forward-KL budget is below $\log[m/(m-1)]$. A missing coordinate can have $H(q) = \log(m-1)$ and $\text{KL}(q||u) = \log[m/(m-1)]$, while $\text{KL}(u||q) = \infty$. For $m = 16$, the entropy threshold is $\log 15$ (about 2.70805 nats), only 0.06454 below $\log 16$. This distinguishes level sets. It does not say that an exact-entropy gradient flow must reach a zero

coordinate; exact entropy can prevent extinction under suitable dynamics. An empirical entropy estimate with missing modes is not an exact value of $H(q)$, and conditional entropy alone does not bound total correct or bank mass.

4.3 When response likelihood yields a key certificate

Fix one prompt and finitely many distinct complete responses e_j , verified to produce keys b_j . Put a normalized distribution w_j on these responses, and let $W_b = \sum_{j:b_j=b} w_j$. Under a fixed decoding law, the response events are disjoint. Extend the exemplar target by zero to every other complete response in the policy alphabet. Execution is then a deterministic map on this full alphabet; unlisted responses remain distinguishable until they are mapped to their keys. Therefore

$$-\sum_j w_j \log \pi(e_j | x) \leq C \implies \text{kl}(W_b, p(b | x)) \leq C - H(w).$$

Indeed, the response-level KL is $C - H(w)$ or smaller, and deterministic execution followed by the binary key indicator is data processing. Thus $p(b | x) \geq \ell_-(W_b, C - H(w))$. Individually, $p(b_j | x) \geq \pi(e_j | x)$ because the complete-response event is contained in its verified-key event.

The likelihood must include EOS or another specified termination event, or a fixed deterministic horizon, and must use the same prompt, temperature, vocabulary mask, token-truncation rule, and verifier as the probability being bounded. A scored prefix is not sufficient: its continuations can produce other keys or invalid outputs. A mean-token log score is not a probability; for a fixed complete length L_j , the sequence probability is $\exp(L_j \bar{\ell}_j)$, not $\exp(\bar{\ell}_j)$. A sum of length-normalized scores first requires normalization: if $a_j = \alpha_j/L_j$, $A = \sum_j a_j$, and $w_j = a_j/A$, then $\sum_j \alpha_j [-\log \pi(e_j)]/L_j = AR_w$. Responses from different prompts do not share one conditional probability alphabet; apply this construction prompt by prompt or specify a joint prompt mixture explicitly. Approximate numerical log scores additionally require controlled numerical error before being called rigorous certificates.

4.4 Finite samples distinguish detection from extinction

At a frozen checkpoint, for a mode specified before sampling, N independent draws give $X \sim \text{Binomial}(N, p_b)$. Exact binomial-tail inversion supplies one-sided confidence bounds [11]. If $X = 0$, the one-sided $(1 - \alpha)$ upper bound is

$$U_0 = 1 - \alpha^{1/N},$$

with lower bound zero. This follows directly from $\mathbb{P}_{p_b}(X = 0) = (1 - p_b)^N$: for any $p_b > U_0$, the all-zero event has probability below α . A full upper-confidence procedure uses the corresponding binomial-tail inversion for nonzero counts as well.

Zero sightings in 32 draws allow p_b as large as 0.08937 at the one-sided 95% level. Simultaneous coverage over 16 prespecified modes by Bonferroni uses $\alpha/16$ and increases that upper bound to 0.16495. At least 299 zero-count draws are needed to obtain a single-mode 95% upper bound below .01; 574 suffice for simultaneous 16-mode coverage. These are probability-threshold statements, not proofs of exact extinction. For example, two sightings in 32 draws give a one-sided 95% lower bound of about .01122 for one prespecified mode.

With repeated inspection of a fixed policy, an anytime-valid confidence sequence can avoid invalid optional stopping [12]. One simple Bernoulli construction uses a fixed beta mixture, with shape parameters $a, b > 0$ and cumulative successes S_t :

$$M_t(p) = \frac{B(S_t + a, t - S_t + b)}{B(a, b)p^{S_t}(1 - p)^{t - S_t}}, \quad C_t = \{p : M_t(p) < 1/\alpha\}.$$

For each true $p \in (0, 1)$, this is an integral of likelihood-ratio martingales, so it is a nonnegative mean-one martingale. The maximal inequality for such martingales gives $\mathbb{P}_p(\exists t : p \notin C_t) \leq \alpha$. Mode-wise error allocations can make this simultaneous over a fixed roster. The argument requires a fixed Bernoulli success probability; pooling across changing training checkpoints does not certify the current checkpoint's probability. Nor do repeated deterministic teacher-forced scores supply Bernoulli trials.

For an unknown catalogue, missing mass is the total probability of as-yet unobserved categories under a fixed iid sampling law, not their number and not evidence that they are extinct. Specialized concentration results address that total mass [13, 14]. They cannot establish the survival of each named banked key, and stationary species formulas cannot be applied directly to a history generated by changing policies. Named-key binomial counts are the direct measurement for per-mode visibility.

5 An exact optimizer comparison from information geometry

Natural gradient on the categorical simplex uses the Fisher metric, also called the Shahshahani metric in evolutionary dynamics [5, Sections 2.3 and 3.1]. This gives a useful comparison with the paper's Euclidean-logit collapse result.

Proposition 8 (Natural-gradient selection and replay). *Start from an interior categorical distribution $p(0)$ with $0 < P(0) < 1$. Let $\Psi \in C^2([0, 1])$ satisfy $c(P) = \Psi'(P) > 0$, let w be a fixed probability vector supported on a nonempty correct-mode bank, and extend it by zero to $\bar{w}$ outside that bank. Exact Fisher natural-gradient ascent of*

$$\mathcal{J}(p) = \Psi(P) + \rho \sum_b w_b \log p_b$$

with $\rho \geq 0$ induces

$$\dot{p}_i = c(P)p_i(\mathbf{1}\{i \in \mathcal{C}\} - P) + \rho(\bar{w}_i - p_i), \quad (1)$$

$$\dot{P} = (1 - P)(c(P)P + \rho), \quad \dot{q}_c = \frac{\rho}{P}(w_c - q_c), \quad (2)$$

where $w_c = 0$ for unbanked correct modes. With $\rho = 0$, correctness tends to one and $q(t) = q(0)$: there is no within-correct-mode collapse. With $\rho > 0$, define $A(t) = \int_0^t \rho/P(s) ds$. Then

$$q(t) = w + (q(0) - w)e^{-A(t)}, \quad A(t) \geq \rho t,$$

and

$$1 - P(t) \leq (1 - P(0))e^{-\rho t}, \quad p_b(t) \geq P(0) \min\{q_b(0), w_b\} > 0$$

for every banked mode and all $t \geq 0$.

Proof. For a smooth objective on the interior simplex, the Fisher natural-gradient vector is $\dot{p}_i = p_i(\partial_i \mathcal{J} - \sum_j p_j \partial_j \mathcal{J})$. Here $\partial_i \mathcal{J} = c(P)\mathbf{1}\{i \in \mathcal{C}\} + \rho \bar{w}_i/p_i$, whose p -weighted mean is $c(P)P + \rho$. This proves (1). Summing over correct categories and differentiating $q_c = p_c/P$ proves (2). Incorrect coordinates obey $\dot{p}_i = -(c(P)P + \rho)p_i$, so remain positive at finite times. Correct coordinates stay positive by the explicit q formula (or are constant in q when $\rho = 0$). Thus the interior calculation applies for all finite times.

For $\rho = 0$, P increases and cannot have a limit below one because $c > 0$ on the compact interval $[P(0), 1]$. For $\rho > 0$, solve the linear q equation and use $P \leq 1$ to obtain $A(t) \geq \rho t$. The differential inequality $\dot{P} \geq \rho(1 - P)$ gives the correctness bound. Each $q_b(t)$ is between $q_b(0)$ and w_b , and $P(t) \geq P(0)$, which gives the probability floor. $\square$

The term $\rho(\bar{w} - p)$ has the form of replenishment from a fixed population distribution. The result is an elementary specialization of the Fisher-gradient formula, not a claim of a new general evolutionary-dynamics theorem. It shows that the update geometry changes both collapse and the strength of retention guarantees. For an incomplete bank it still sends unbanked correct mass to zero. It supplies no guarantee for approximate natural gradient, neural Fisher matrices, PPO, or AdamW, and the time variable is an idealized optimization clock rather than elapsed compute time.

6 What is ready to use in the paper

The sharp KL certificate and the exact-entropy distinction directly strengthen the existing retention discussion. The optimizer theorem extends the fixed joint-potential argument to controlled stochastic updates. The admission propositions identify sufficient conditions for a discovery-to-retention claim, including the effects of finite capacity and bank changes. The natural-gradient calculation is an optional mechanism comparison that makes the role of optimizer geometry particularly explicit.

Applying these statements to the released neural runs still requires their premises to be measured or established. In particular, exact complete-response likelihood and finite categorical replay cross entropy must not be confused with mean-token scores. A theorem about probability retention cannot substitute for observations that individual stored modes survived the actual training trajectory. The registered E121 study remains prospective at this audit: no per-mode telemetry was available in its five registered run directories. The accompanying survival audit identifies analysis work needed before those records can support the registered claims.

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